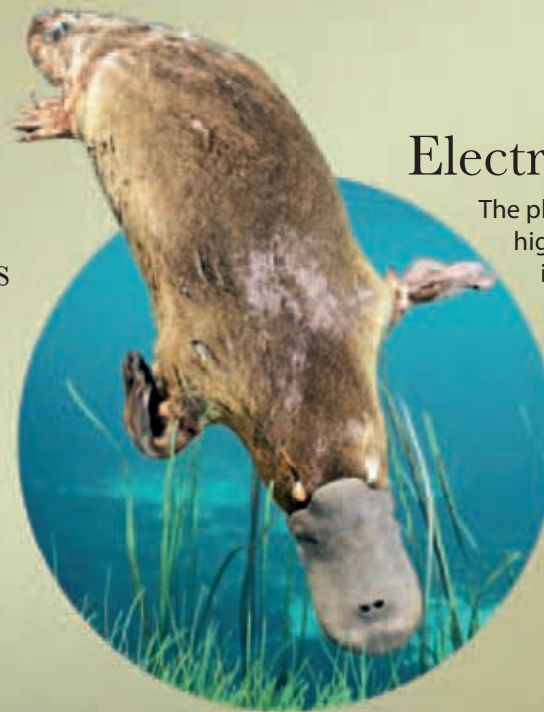


Special senses

Many animals sense things that we cannot and that are hard for us to imagine. On land, some snakes “see” infrared radiation, which lets them hunt warm-blooded animals after dark. In water, platypuses and sharks sense the faint electric fields produced by their prey—using a special sense known as electroreception. But special senses are not used only to find food. A wide variety of animals, including turtles, birds, and bats, can sense the strength and direction of Earth’s magnetic field. When they migrate, their magnetic sense can help to guide them to their destination.



Electric inspection

The platypus’s rubbery bill is highly sensitive to touch, but it also has around 40,000 tiny sensors that detect electric fields. When the platypus is hunting, it sweeps its bill from side to side, which lets it sense the electric fields of small animals buried in the riverbed mud. Once it has found the food, it uses its bill to dig it out quickly.

Sensing heat

All warm objects give off infrared radiation—a form of energy that carries heat. Rattlesnakes, pythons, and boas can detect it at extremely low levels, through heat-sensitive pits in their faces.

Using its pits, this green tree python can strike mammals or birds in total darkness, as long as they are against a colder background. Rattlesnakes have two pits, but pythons and boas have several along the sides of their mouths.

Sensory signals from eye are processed by same part of brain as signals from pits

Wedge-shaped pit lined with heat-sensitive nerves